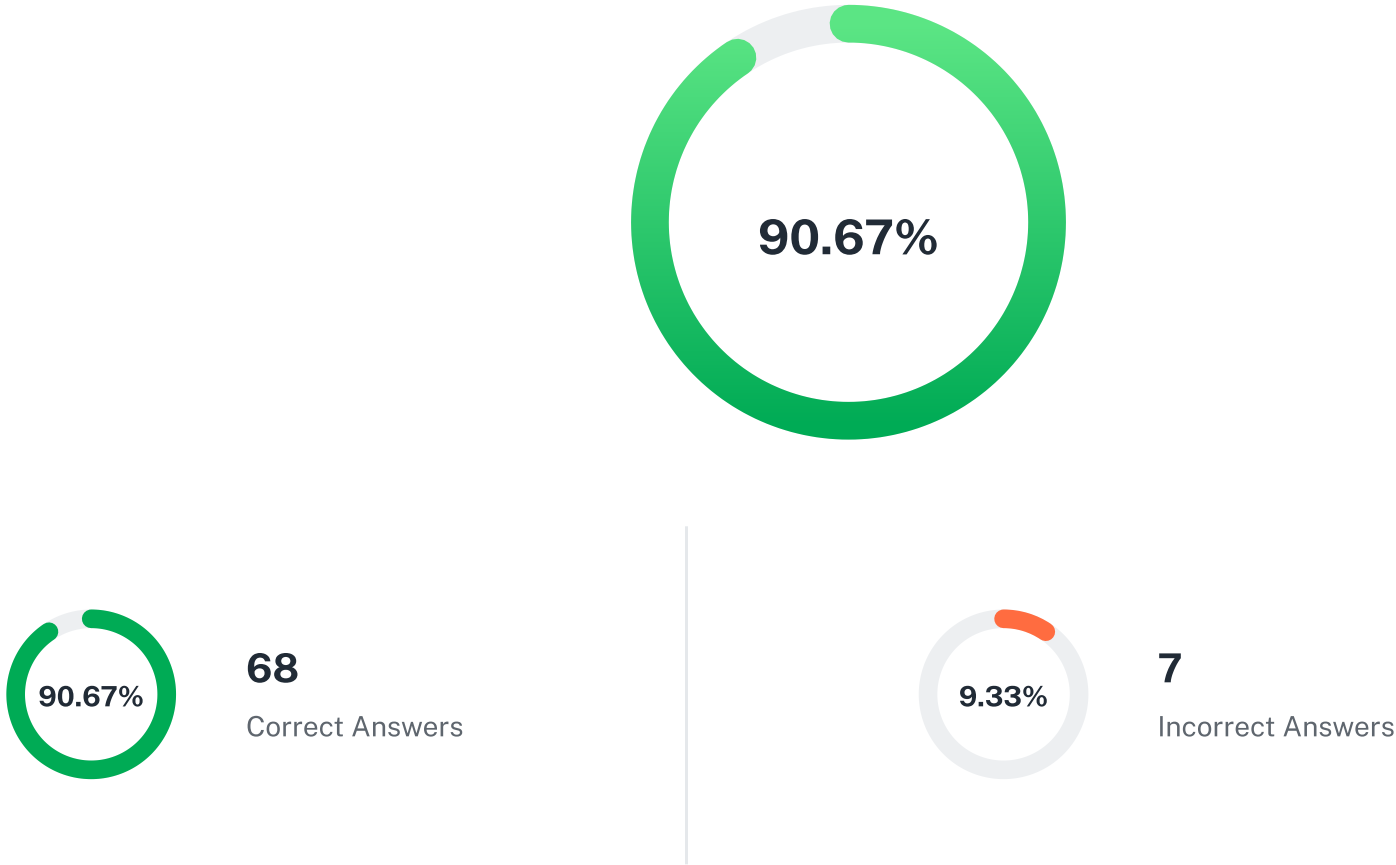


Practice Test Review

Review 82feba17-5692-4d0c-a887-d8eb80c...

NVIDIA : NCP-AAI

Overall Test Result



The practice test includes only multiple-choice questions (MCQs) due to development limitations. To achieve a good score, you need to review the premium PDF.

All Correct Only Incorrect Only

Question: 1

CertyIQ

A company is deploying an AI-powered customer support agent that integrates external APIs and handles a wide range of customer inputs dynamically. Which of the following strategies are appropriate when designing an AI agent for dynamic conversation management and external system interaction? (Choose two.)

- ☒ Integrating a feedback loop from user interactions to iteratively improve agent behavior. ✓
- ☐ Using rule-based logic as the primary framework to maintain consistency in agent decisions.
- ☒ Implementing retry logic for API failures to ensure robustness in external communications. ✓
- ☐ Preferring hardcoded responses for frequent queries to deliver reliable and low-latency answers.

Show Explanation

Question: 2

CertyIQ

When implementing inter-agent communication for a distributed agentic system running across multiple NVIDIA GPU nodes, which message routing pattern provides the best balance of reliability and performance?

- ☐ Database-based message queuing with polling
- ☐ Direct TCP connections between all agent pairs
- ☒ Event-driven message routing with distributed broker clusters ✓
- ☐ Centralized message broker with topic-based routing

Show Explanation

Question: 3

CertyIQ

Which two orchestration methods are MOST suitable for implementing complex agentic workflows that require both external data access and specialized task delegation? (Choose two.)

- ☒ Agentic orchestration with specialized expert system delegation ✓
- ☐ Prompt chaining to accomplish state management
- ☐ Manual workflow coordination without automation
- ☒ Retrieval-based orchestration for external data ✓
- ☐ Static rule-based routing with predefined pathways

Show Explanation

Question: 4

CertyIQ

Which two coordination patterns are MOST effective for implementing a multi-agent system where agents have different specializations (Research Analyst, Content Writer, Quality Validator)?

- ☒ Sequential pipeline coordination with crew-based structured handoffs ✓
- ☐ Peer-to-peer coordination with consensus mechanisms
- ☐ Random task distribution with load balancing
- ☒ Hierarchical coordination with crew-based task delegation ✓

Show Explanation

Question: 5

CertyIQ

You are designing a virtual assistant that helps users check weather updates via external APIs. During testing, the agent frequently calls the incorrect tools, often hallucinating endpoints or returning incorrect formats. You suspect the prompt structure might be the root cause of these failures.

Which prompt design best supports consistent tool invocation in this agent?

- ☐ Rely on the agent’s internal knowledge to infer tool usage
- ☐ Include tool names in natural language but without parameter examples
- ☐ Provide only a generic system instruction with no examples
- ☒ Use structured prompt templates with few-shot tool usage examples ✓

Show Explanation

Question: 6



What NVIDIA framework can be used to train a better agent?

- ☒ NeMo-RL ✓
- ☐ NeMo Guardrails
- ☐ TensorRT-LLM

Show Explanation

Question: 7



You are evaluating your RAG pipeline. You notice that the LLM-as-a-Judge consistently assigns high similarity scores to responses that contain irrelevant information.

What should you investigate as the most likely potential cause with the least development effort?

- ☐ The temperature setting used by the LLM during response generation.
- ☐ The size of the knowledge base used to power the RAG pipeline.
- ☐ The quality of the synthetic questions used for evaluation.
- ☒ The prompt used to instruct the LLM-as-a-Judge to assess the response. ✓

Show Explanation

Question: 8



A financial services agentic AI is being used to automate initial customer onboarding. The agent is completing the process efficiently and accurately, but reviews of its conversations reveal it often uses overly formal and complex language that confuses customers.

Which type of evaluation is best suited to address this issue?

- ☒ Controlled user testing sessions to collect user feedback on the clarity and tone of responses ✓
- ☐ Compliance review of the agent’s access to regulatory guidelines and policy documentation
- ☐ Continuous user feedback collection, specifically gathering subjective assessments of the agent’s communication style
- ☐ Statistical analysis of the agent’s decision-making patterns to detect overly formal and complex response choices

Show Explanation

Question: 9



When analyzing an agent’s failure to complete multi-step financial analysis tasks, which evaluation approach best identifies prompt engineering improvements needed for reliable task decomposition and execution?

- ☒ Implement systematic prompt testing with chain-of-thought reasoning templates, step-by-step decomposition analysis, and success rate tracking across tasks of varying complexity. ✓
- ☐ Focus primarily on response speed optimization as a primary focus over reasoning quality, step completion accuracy, and prompt clarity for complex analytical requirements.
- ☐ Test only final output accuracy as this will automatically include intermediate reasoning steps, decomposition quality, and prompt structure effectiveness for complex workflows.
- ☐ Rely on generic prompt templates which are by default already optimized for general use, instead of tailoring them to financial terminology, calculation needs, or specialized multi-step analysis patterns.

Show Explanation

Question: 10

CertyIQ

When analyzing suboptimal agent response quality after deployment, which parameter tuning evaluation methods effectively identify the optimal configuration adjustments? (Choose two.)

- ☒ Design ablation studies systematically varying individual parameters while holding others constant to isolate each parameter’s impact on agent behavior and performance. ✓
- ☐ Apply identical parameter settings across all agent types and tasks, promoting consistency and simplifying comparison across different use cases.
- ☒ Implement A/B testing frameworks comparing temperature, top-k, and top-p variations while measuring task-specific quality metrics and user satisfaction scores. ✓
- ☐ Use production traffic directly for parameter experiments, enabling real-world insights and faster identification of impactful settings.
- ☐ Randomly adjust all parameters simultaneously, allowing for broader exploration of the parameter space in a shorter time frame.

Show Explanation

Question: 11

CertyIQ

You’re employing an LLM to automate the generation of email responses for a customer service team. The generated responses frequently miss the mark, failing to address the customer’s underlying concerns. What’s the most crucial element to add to the prompt to enhance the quality of the email responses?

- ☒ Instructing the LLM with a detailed prompt containing instructions on how to format and compose the response in an easy-to-understand structure. ✓
- ☐ Instructing the LLM to use a simple template for all email replies before generating a response.
- ☐ Instructing the LLM to “understand the customer’s issue” before generating a response.
- ☐ Instructing the LLM to provide a response that “is the most helpful” before generating a response.

Show Explanation

Question: 12

CertyIQ

A team is evaluating multiple versions of an AI agent designed for customer support. They want to identify which version completes tasks more efficiently, responds accurately, and improves over time using user feedback. Which practice is most important to ensure continuous refinement and optimal performance of the AI agent?

- ☐ Comparing agents on isolated tasks without standardized benchmarking pipelines
- ☐ Relying solely on offline benchmarks without incorporating live user feedback during tuning
- ☒ Implementing an evaluation framework that quantifies task efficiency and incorporates human-in-the-loop feedback ✓
- ☐ Tuning model parameters once before deployment to maximize initial accuracy

Show Explanation

Question: 13

CertyIQ

Your agent is generating inconsistent and contradictory statements. Which approach would be most suitable to improve the agent’s output?

- ☒ Employing Reflexion ✓
- ☐ Increasing the number of generated plans
- ☐ Using Decomposition-First Planning
- ☐ Decreasing the length of prompts

Show Explanation

Question: 14

CertyIQ

You’re utilizing an LLM to translate complex technical documentation into multiple languages. The translations often lack nuance and fail to capture the original intent. What’s the most effective strategy for improving the quality of the translations?

- ☒ Providing the LLM with a glossary of key terms, concepts in all languages and the dataset of previously translated text. ✓
- ☐ Training the LLM on a dataset of translated texts.
- ☐ Providing the LLM with guidance to “translate the documents” without additional guidance, so it can use trained knowledge.
- ☐ Providing the LLM with guidance to translate “with high accuracy” without additional guidance, so it can use trained knowledge.

Show Explanation

Question: 15

CertyIQ

A technology startup is preparing to launch an AI agent platform to serve clients with unpredictable usage patterns. They face periods of high user activity and low demand, so their deployment approach must minimize wasted resources during slow times and automatically allocate more resources during busy periods –all while keeping operational costs reasonable. Given these requirements, which deployment strategy most effectively ensures both cost-effectiveness and adaptability for scaling agentic AI systems?

- ☐ Scheduling periodic manual reviews to increase or decrease infrastructure based on predicted user numbers
- ☐ Monitoring system logs for usage patterns and making infrastructure changes after monthly analysis
- ☐ Using fixed-size virtual machine clusters to guarantee consistent resource allocation at all times
- ☒ Implementing autoscaling policies in a container orchestration environment to automatically adjust resources according to workload changes ✓

Show Explanation

Question: 16

CertyIQ

You are deploying a multi-agent customer-support system on Kubernetes using NVIDIA GPU nodes and Triton Inference Server. Traffic spikes during product launches. You need <100ms response times, zero downtime, automatic GPU scaling, and full monitoring. Which deployment setup best achieves cost-effective, reliable, low-latency scaling?

- ☐ Set up one mixed GPU node pool with Cluster Autoscaler min=0, scale by network throughput, monitor via metrics-server and logs, and skip readiness probes for fast startup.
- ☐ Place GPU pods on on-demand nodes in one zone, disable Cluster Autoscaler, run a fixed pod count for bursts, scale on CPU usage, and monitor with default health checks.
- ☒ Deploy GPU pods in a node pool spanning all zones, mix GPU types, enable Cluster and Horizontal Pod Autoscalers using Prometheus GPU and latency metrics, and monitor with NVIDIA DCGM and Grafana. ✓
- ☐ Use spot-instance node pools across zones, enable Cluster Autoscaler with capped nodes, scale on memory usage, and monitor with logs and cluster events.

Show Explanation

Question: 17

CertyIQ

Which two deployment patterns are MOST suitable for scaling agentic workloads on NVIDIA Infrastructure? (Choose two.)

- ☐ Bare metal deployment with manual resource allocation
- ☐ Static virtual machine deployment with fixed resources
- ☐ Serverless deployment without GPU acceleration
- ☒ Containerized deployment with NIM (NVIDIA Inference Microservices) ✓
- ☒ Kubernetes orchestration with Horizontal Pod Autoscaling (HPA) ✓

Show Explanation

Question: 18

CertyIQ

A company operates agent-based workloads in multiple data centers. They want to minimize latency for users in different regions, maintain continuous service during infrastructure upgrades, and keep operational costs predictable. Which deployment practice best supports low-latency, resilient, and cost-efficient agent operations at scale?

- ☐ Schedule regular agent downtime for system updates and operational recalibration.
- ☒ Implement geo-distributed deployments with rolling updates and resource usage monitoring. ✓
- ☐ Prioritize high-performance GPUs for all agents in geo-distributed deployments.
- ☐ Apply static infrastructure allocation with centralized resource usage monitoring at a single data center.

Show Explanation

Question: 19

CertyIQ

When implementing stateful orchestration for agentic workflows using LangGraph, which memory management approach provides the best balance of performance and context retention?

- ☐ Store complete conversation history in memory with periodic database syncing
- ☐ Implement rolling window memory with fixed conversation length limits
- ☒ Use session-ID based checkpointer with user-defined schema for selective state persistence ✓

Show Explanation

Question: 20

CertyIQ

An AI Engineer at an automotive company is developing an inventory restocking assistant for parts that must plan reordering of parts over multiple days, factoring in stock levels, predicted demand, and supplier lead time. Which approach best equips the agent for sequential decision-making?

- ☐ Reinforcement learning sequence model using only a custom PyTorch Decision Transformer
- ☐ Rule-based reorder strategy with fixed thresholds implemented via NVIDIA Triton Inference Server
- ☐ Hybrid supervised/RL-trained model using NeMo-Aligner for policy alignment
- ☒ Reinforcement learning sequence model such as NVIDIA'S NeMo-RL framework ✓

Show Explanation

Question: 21

CertyIQ

An AI engineer at an oil and gas company is designing a multi-agent AI system to support drilling operations. Different agents are responsible for subsurface modeling, risk analysis, and resource allocation. These agents must share operational context, reason through interdependent planning steps, and justify their collaborative decisions using structured, transparent logic. The architecture must support memory persistence, sequential decision-making and chain-of-thought prompting across agents. Which implementation best supports this design?

- ☒ Orchestrate NeMo agents via Triton, use vector memory for shared context, ReAct planning, and NeMo Guardrails for reasoning. ✓
- ☐ Use stateless LLM endpoints behind an API gateway and pass shared prompts across agents to simulate context and reasoning.
- ☐ Use LangChain to coordinate third-party agent APIs and store shared information in external memory, with logic encoded in static prompt chains.
- ☐ Fine-tune separate NeMo models for each agent role using LoRA, with pre-scripted action flows deployed via TensorRT for latency reduction.

Show Explanation

Question: 22

CertyIQ

You are creating a virtual assistant agent that needs to handle an increasingly wide range of tasks over an extended period. What is the primary benefit of combining external storage (like RAG) with fine-tuning (embodied memory) in this context?

- ☒ To enhance long-term reasoning capabilities and adaptability ✓
- ☐ To accelerate the agent's initial response time
- ☐ To ensure the agent doesn't make any errors
- ☐ To eliminate the need for external knowledge

Show Explanation

Question: 23

CertyIQ

A development team is building an AI agent capable of autonomously planning and executing multi-step tasks while retaining context and learning from past interactions. Which practice is most important to enable the agent to effectively manage long-term memory and complex tasks?

- ☒ Implement memory mechanisms for context retention and apply chain-of-thought prompts to enhance reasoning. ✓
- ☐ Use basic rule-based decision methods that emphasize fast responses over adaptive planning.
- ☐ Apply short-term memory approaches that handle each interaction independently of previous ones.
- ☐ Reduce planning features and memory management to keep the system streamlined.

Show Explanation

Question: 24

CertyIQ

When analyzing memory-related performance degradation in agents handling extended customer support sessions, which evaluation methods effectively identify optimization opportunities for context retention? (Choose two.)

- ☐ Clear memory after each interaction and reset session state, removing historical context needed for personalized tasks to identify optimization opportunities.
- ☒ Profile memory access patterns by measuring retrieval latency, relevance scoring accuracy, and storage efficiency while monitoring context window utilization to identify optimization opportunities. ✓
- ☐ Use fixed memory allocation including all conversation types, topic changes, and user needs, allowing adaptive-free observation of interaction patterns to identify optimization opportunities.
- ☒ Implement sliding window analysis comparing context compression strategies, summarization quality, and information preservation rates across varying conversation lengths to identify optimization opportunities. ✓
- ☐ Store all conversation history including all interactions, allowing adaptive-free observation of data to identify optimization opportunities.

Show Explanation

Question: 25

CertyIQ

A team is designing an AI assistant that helps users with travel planning. The assistant should remember user preferences, build personalized itineraries, and update plans when users provide new requirements. Which approach best equips the AI assistant to provide personalized and adaptive travel recommendations?

- ☐ Using a single-step question-answering system enhanced with session-level keyword tracking to improve relevance during ongoing interactions.
- ☐ Designing the assistant to handle each user request independently, while using implicit signals within each session to suggest relevant options.
- ☒ Engineering multi-step reasoning frameworks with persistent memory systems to store and utilize user preferences. ✓
- ☐ Providing the same set of travel options to every user but sorting them based on recent popular destinations.

Show Explanation

Question: 26

CertyIQ

Implement Memory Systems for Contextual Awareness
An enterprise AI system needs to maintain contextual information over multiple interactions with users. Which memory implementation approach would be MOST effective for managing both immediate context and long-term historical interactions within an agentic workflow?

- ☐ Rely predominantly on the context window of the base LLM model to store all historical interactions with minimal external memory supplementation.
- ☒ Implement a hybrid memory system with short-term memory for immediate context and a vector database for long-term memory with semantic retrieval capabilities. ✓
- ☐ Use a static prompt template with fixed context for all interactions, thereby providing memory information in that form across conversation sessions.
- ☐ Store all user interactions in a simple key-value database which will by default provide organization and retrieval strategy for historical context management.

Show Explanation

Question: 27

CertyIQ

A company is building an AI agent that must retrieve information from large document collections and client databases in real time. The team wants to ensure fast, accurate retrieval and maintain high data quality. Which approach best supports efficient knowledge integration and effective data handling for such an agent?

- ☐ Using traditional relational databases because they don't need specialized retrieval mechanisms for all data queries
- ☐ Integrating client data sources as they already incorporate data quality checks or augmentation to speed up deployment
- ☐ Relying on pre-trained models instead of connecting to external knowledge sources during inference
- ☒ Implementing retrieval-augmented generation (RAG) pipelines combined with vector databases to accelerate access to relevant information ✓

Show Explanation

Question: 28

CertyIQ

What is RAG Fusion primarily designed to achieve?

- ☐ Creating a separate, dedicated database for storing all the retrieved chunks.
- ☐ Minimizing the need for retrieval, allowing the LLM to generate responses directly from its internal knowledge.
- ☒ Blending information from multiple retrieved chunks into a single response generated by the LLM. ✓
- ☐ Automatically translating and integrating all retrieved chunks into a single language.

Show Explanation

Question: 29

CertyIQ

Your agent's primary task is to collect data from a device management platform API. This API frequently returns data in JSON format, but the underlying schemas can evolve without detailed documentation. What's the most effective strategy for the agent to maintain consistent access to the data, despite potential schema changes?

- ☐ Building a JSON schema validator that constantly monitors and enforces the current API schema.
- ☐ Manually inspecting the API response's structure and updating the agent's code when a change is detected.
- ☐ Building a fixed data extraction method based on the API's historical response patterns.
- ☒ Building a flexible data mapping layer that can adapt to changes in the API response structure. ✓

Show Explanation

Question: 30

CertyIQ

You are improving your Retrieval-Augmented Generation (RAG) pipeline, and you want to exploit the properties of the embedding model to improve the solutions. Within a RAG system, why is reranking retrieved chunks a crucial step after initial retrieval?

- ☐ Reranking improves retrieval accuracy, regardless of the initial retrieval method.
- ☐ Reranking's purpose is to reduce the overall length of the response generated by the LLM.
- ☒ Reranking allows for prioritizing chunks based on relevance to the semantic intent of the query. ✓
- ☐ Reranking primarily focuses on optimizing the visual presentation of the retrieved chunks.

Show Explanation

Question: 31

CertyIQ

In your RAG deployment, you've identified a performance bottleneck in the retrieval phase – specifically, the time it takes to access the vector database.

Which of the following optimization strategies is most aligned with micro-service best practices, considering your RAG architecture?

- ☐ Implement a “cache-and-check” mechanism where the retrieval microservice immediately returns the first matching chunk, regardless of relevance.
- ☐ Increase the size of the LLM model itself, because it will automatically accelerate the overall response time.
- ☒ Introduce a dedicated service responsible solely for querying the vector database and returning relevant chunks. ✓
- ☐ Optimize the LLM prompt to be shorter and more concise, significantly reducing the computational load.

Show Explanation

Question: 32

CertyIQ

You are designing the architecture for a RAG (Retrieval-Augmented Generation) system, and you are concerned about ensuring data freshness and minimizing latency.
Which of the following is the most important consideration when designing the architecture?

- ☐ Employing a consolidated architecture with a large service handling all data retrieval and LLM interaction. This ensures consistent performance and simplifies debugging.
- ☐ Using a synchronous, block-level approach, where the LLM continuously monitors the database for updates and retrieves the entire dataset with each prompt.
- ☒ Implementing a single, centralized database for all data, updated with a synchronous polling mechanism for the LLM to retrieve the latest information. X
- ☐ Use a loosely coupled, event-driven micro-service architecture where separate services handle data indexing, retrieval, and LLM prompting. ✓

Show Explanation

Question: 33

CertyIQ

Which two optimization strategies are MOST effective for improving agent performance on NVIDIA GPU infrastructure? (Choose two.)

- ☒ Using multi-GPU coordination to distribute workloads, enabling higher throughput and efficiency for scaling agent tasks. ✓
- ☒ Applying TensorRT-LLM optimizations to reduce inference latency by improving kernel efficiency and memory usage. ✓
- ☐ Expanding GPU memory capacity to support larger models, assuming this alone guarantees meaningful performance improvements.
- ☐ Manually tuning kernel launch parameters to optimize individual operations while overlooking overall pipeline performance dynamics.

Show Explanation

Question: 34

CertyIQ

A healthcare AI company is deploying diagnostic agents that process medical imaging and patient data. The system must deliver consistent sub-100ms inference times for critical diagnoses while supporting deployment across multiple hospital sites with different NVIDIA GPU configurations (from RTX 6000 workstations to DGX systems). The agents need to maintain high accuracy while being portable across different hardware environments and capable of running efficiently on various GPU memory configurations.
Which optimization strategy would deliver the BEST performance improvements while maintaining deployment flexibility across diverse NVIDIA hardware configurations?

- ☐ Deploy agents with NVIDIA CUDA-optimized Docker containers using a sequential inference architecture that processes each layer individually with GPU-to-CPU memory transfers between operations to avoid memory issues.
- ☒ Deploy agents using NVIDIA NIM containers with CPU-optimized inference to avoid GPU memory constraints and ensure consistent performance across different hospital infrastructure configurations. X

- ☐ Deploy models using NVIDIA TensorRT optimization in their original FP32 precision format without any quantization or memory optimization, requiring 32GB+ GPU memory across all deployment sites.
- ☒ Deploy agents using model optimizations with post-training quantization with Nvidia NIM deployment for portable performance across different GPU platforms and memory configurations. ✓

Show Explanation

Question: 35

CertyIQ

You are tasked with deploying a multi-modal agentic system that must respond to user queries with minimal latency while maintaining guardrails for safe and context-aware interactions. Which of the following configurations best leverages NVIDIA's AI stack to meet these requirements?

- ☒ Integrate NeMo Guardrails, configure NIM microservices for optimized inference, use TensorRT-LLM for deployment, and profile the system using Triton Inference Server with multi-modal support. ✓
- ☐ Integrate NeMo Guardrails, use Omniverse to generate synthetic data, configure NIM microservices for optimized inference, use TensorRT-LLM for deployment, and profile the system using NeMo Agent Toolkit for multi-modal support.
- ☐ Use NeMo Guardrails for safety, deploy the model with Triton Inference Server using default settings, and rely on hardware accelerators like GPU/TPU inference for cost efficiency.
- ☐ Use NIM microservices for deployment, optionally use NeMo Guardrails unless one wants to minimize the inference overhead.

Show Explanation

Question: 36

CertyIQ

Your team notices a spike in failed tool calls from a deployed workflow agent after a recent API schema update. The agent still returns outputs, but many are irrelevant or incomplete. Which maintenance task should be prioritized to restore accurate behavior?

- ☐ Reset the agent's long-term memory and reinitialize logs.
- ☒ Update the tool function specifications and re-test action sequences. ✓
- ☐ Increase model temperature to encourage tool exploration.
- ☐ Reduce tool retrieval vector similarity threshold to broaden context.

Show Explanation

Question: 37

CertyIQ

A recently deployed agent sometimes outputs empty responses under heavy system load. Which system-level signal is most useful for diagnosing this issue?

- ☐ Number of tool function arguments returned per query
- ☐ Retrieval similarity thresholds in vector search
- ☒ GPU memory utilization and server-side inference logs ✓
- ☐ Prompt injection detection rate over time

Show Explanation

Question: 38

CertyIQ

An AI architect at a national healthcare provider is maintaining an agentic AI system. The system must monitor model and system performance in real time, raise alerts on failures or anomalies, manage version control and rollback of diagnostic models, and provide transparent insight into agent behavior during patient care workflows. Which operational approach best supports these requirements using the NVIDIA AI stack?

- ☐ Containerize each agent in NIM with basic health checks running on cron jobs, and manage version rollback by swapping prebuilt container images.
- ☐ Optimize all models with TensorRT and use periodic manual log reviews and NVIDIA shell scripts for detecting service anomalies and managing rollback.
- ☒ Deploy agent models on NVIDIA Triton Inference Server with Prometheus and Grafana for performance alerting, and manage model lifecycle via NGC and the Triton model repository. ✓
- ☐ Expose agents as stateless NVIDIA API endpoints and monitor activity through application logs, with model versions tracked in a Git-based script repository.

Show Explanation

Question: 39

CertyIQ

When evaluating a customer service agent’s resilience to API failures and network issues, which analysis methods effectively identify weaknesses in error handling and retry mechanisms? (Choose two.)

- ☒ Analyze retry logic for exponential backoff patterns, retry limits, and circuit breaker integration to prevent cascading failures in distributed systems. ✓
- ☐ Implement retry mechanisms that standardize recovery attempts across scenarios, emphasizing consistency in handling errors.
- ☐ Use fixed retry intervals to avoid the pitfalls of dynamic tuning, keeping retry timing consistent across different error conditions.
- ☐ Test under normal network conditions to establish baseline behavior, comparing results against production performance during degraded service scenarios.
- ☒ Conduct failure injection testing with varied error types (timeouts, rate limits, malformed responses) while monitoring recovery patterns and fallback behavior. ✓

Show Explanation

Question: 40

CertyIQ

Which two error handling strategies are MOST important for maintaining agent reliability in production environments? (Choose two.)

- ☒ Circuit breaker patterns for external service calls ✓
- ☐ Immediate failure propagation to users with verbose logging
- ☒ Automatic retry with exponential backoff for transient failures ✓
- ☐ Immediate system shutdown for error handling

Show Explanation

Question: 41

CertyIQ

A customer service agent sometimes fails to complete multi-step workflows when APIs respond slowly or inconsistently. Which approach most effectively increases robustness when working with unreliable APIs?

- ☐ Restrict available tools to reduce decision complexity
- ☒ Add retries with exponential backoff and set request timeouts ✓
- ☐ Cache recent API results to limit unnecessary repeated calls

☐ Adjust generation parameters to produce more predictable responses

Show Explanation

Question: 42

CertyIQ

After deploying a financial assistant agent, users report occasional inconsistencies in how transactions are categorized. What is the best first step for diagnosing the issue?

- ☐ Review and modify prompt temperature to enhance precision
- ☒ Review and retrain the model with more financial datasets X
- ☐ Implement agent memory reset after each session
- ☐ Review tool call inputs and outputs in recent session logs ✓

Show Explanation

Question: 43

CertyIQ

Which two validation approaches are MOST critical for ensuring agent reliability in production deployments? (Choose two.)

- ☒ User satisfaction surveys as the primary quality metric X
- ☐ Performance testing during development phases
- ☐ Structured output validation with Pydantic schemas ✓
- ☐ Random sampling of agent interactions for manual review
- ☒ Automated consistency checking across multiple agent runs ✓

Show Explanation

Question: 44

CertyIQ

Your team has built an agent using LangChain and needs to implement guardrails for deployment in a production environment. Which approach represents the MOST effective integration of NVIDIA NeMo Guardrails?

- ☐ Rebuild the agent using only NeMo Guardrails, thereby reconstructing the LangChain implementation with enhanced safety controls and production-ready guardrail integration.
- ☒ Wrap the LangChain agent with NeMo Guardrails configuration while maintaining the existing workflow architecture and preserving current development investments. ✓
- ☐ Configure input filtering to address safety requirements, integrating guardrail mechanisms focused on data validation and moderation within the current framework.
- ☐ Run the LangChain agent in parallel with NeMo Guardrails, allowing comparison of outputs between systems for comprehensive safety validation and performance optimization.

Show Explanation

Question: 45

CertyIQ

Your team has deployed a generative agent for internal HR use, including summarizing candidate resumes and suggesting interview questions. After deployment, you’ve noticed that the model occasionally associates certain names or genders with particular roles. Which mitigation strategy is the most effective and scalable for reducing this type of bias in agent outputs?

- ☐ Adjust system prompts to explicitly instruct the agent to avoid assumptions based on demographic features
- ☐ Randomly replace names in prompts to reduce identity correlation
- ☐ Add more training examples to the training dataset and re-train the model
- ☒ Implement guardrails to prevent outputs referencing protected attributes ✓

Show Explanation

Question: 46

CertyIQ

When analyzing safety violations in a financial advisory agent that uses NeMo Guardrails, which evaluation approach best identifies gaps in guardrail coverage?

- ☐ Apply keyword- and rule-based validation methods to confirm compliance with policy terms and common risk conditions.
- ☒ Analyze violation patterns, test adversarial prompts, measure guardrail activation, and align policies with observed failures. ✓
- ☐ Conduct functional testing with representative user inputs to verify policy enforcement in typical usage scenarios.
- ☐ Monitor overall guardrail activations and system logs to assess operational behavior across different interaction types.

Show Explanation

Question: 47

CertyIQ

You’re deploying a healthcare-focused agentic AI system that helps doctors make treatment recommendations based on patient records. The agent’s reasoning is not exposed to users, and its decisions sometimes differ from clinical guidelines. What safety and compliance mechanisms should be in place? (Choose two.)

- ☒ Allow overrides by human doctors to maintain accountability ✓
- ☒ Require model explainability or traceability for all outputs ✓
- ☐ Prioritize autonomous speed of decision over explainability
- ☐ Exempt the model from compliance if it improves outcomes
- ☐ Obfuscate decision logic to protect proprietary methods

Show Explanation

Question: 48

CertyIQ

An AI Engineer is analyzing a production agentic AI system’s compliance with responsible AI standards. Which evaluation approaches effectively identify potential safety vulnerabilities and ethical risks in multi-agent workflows? (Choose two.)

- ☐ Emphasize latency metrics and throughput performance as key evaluation factors for safety vulnerabilities, providing a baseline for operational measures and resource allocation.
- ☒ Implement comprehensive audit trails using NVIDIA NeMo Guardrails with semantic similarity checks, tracking agent decisions across conversation flows and evaluating policy violations through automated compliance scoring. ✓
- ☐ Use user feedback as a primary signal for risk identification, emphasizing post-deployment observations and qualitative experience reports alongside operational monitoring.
- ☒ Deploy multi-layered evaluation combining bias detection metrics (demographic parity, equalized odds) with adversarial testing to probe agent responses for harmful outputs across diverse user populations ✓

Show Explanation

Question: 49



You are deploying an AI-driven applicant-screening agent that analyzes candidate resumes and social-media data to recommend top applicants. Due to anti-discrimination laws and corporate policy, the system must mitigate bias against protected groups, maintain an audit trail of decisions, and comply with GDPR (including data minimization and explicit consent). Which of the following strategies is most effective for ensuring your screening agent both mitigates bias in its recommendations and complies with data-privacy regulations?

- ☐ Perform a post-deployment GDPR and bias audit and process raw personal data as received.
- ☒ Pseudonymize protected attributes, implement fairness-aware debiasing, maintain an audit trail, and enforce GDPR data-minimization and consent. ✓
- ☐ Encrypt all candidate data at rest and in transit, remove protected attributes from analysis, and conduct manual bias checks on recommendations.
- ☐ Exclude gender and ethnicity fields during training, use a generic privacy policy for consent, and do not maintain audit logs or apply targeted debiasing.

Show Explanation

Question: 50



A health assistant agent has been running on production environment for several weeks. The compliance team wants to audit how personal health data has been processed. Which operational feature supports this requirement?

- ☐ Adding more prompt examples to clarify privacy rules
- ☐ Masking all output with a profanity and PII detector
- ☐ Increasing model temperature for diverse interpretations
- ☒ Enabling full session logging with audit trail metadata ✓

Show Explanation

Question: 51



You are designing an AI-powered drafting assistant for contract lawyers. The assistant suggests standard clauses and highlights potential risks based on past agreements. Senior attorneys must review, accept, modify, or reject each suggestion, see why a clause was recommended, and provide feedback to help improve the assistant. Which design feature is most critical for enabling effective human-in-the-loop oversight, transparency, and trust?

- ☐ Display suggested clauses with links to additional details about provenance and risk highlighting in a side panel, allowing users to access more context as needed.
- ☐ Insert suggested clauses into the draft and highlight changes for review at the end, inviting users to provide detailed feedback on clauses they wish to flag for improvement.
- ☐ Present batch “accept all” or “reject all” controls for suggested clauses, with explanations and feedback collected in a summary report after draft review.
- ☒ Show inline “why” explanations for each suggestion, highlight precedent and risk factors, and include accept/modify/reject controls with immediate feedback capture for model refinement. ✓

Show Explanation

Question: 52



A logistics company is implementing an agentic AI system for supply chain optimization that manages inventory levels, predicts demand, and automatically reorders supplies across multiple warehouses. Supply chain managers need to monitor AI decisions, understand the reasoning behind inventory recommendations, and intervene when business conditions change rapidly. The system must present complex data analytics in an intuitive way that enables quick decision-making while providing detailed insights when needed. Managers have varying levels of technical expertise and

need interfaces that support both high-level oversight and detailed analysis.
Which user interface design approach would BEST support effective human oversight of this complex multi-agent supply chain system?

- ☐ Develop a comprehensive dashboard with AI decision summaries, drill-down access to underlying data sets, and segmented performance metrics to enable targeted analysis of supply chain operations.
- ☐ Create separate specialized interfaces tailored to specific user roles, allowing managers to view AI-driven recommendations with drill-down options for role-specific details, but without a unified interface for cross-role collaboration.
- ☒ Create a layered interface featuring intuitive summaries, drill-down capabilities for detailed analysis, contextual explanations of AI decisions, and clear intervention controls with impact visualization and decision support tools. ✓
- ☐ Create a streamlined interface presenting only high-level AI decisions and simplified recommendations, with drill-down views limited to basic historical trends for quick reference.

Show Explanation

Question: 53

CertyIQ

When analyzing a customer service agentic system’s performance degradation over time, which evaluation approach most effectively identifies opportunities for human-in-the-loop intervention to improve agent decision-making transparency and user trust?

- ☐ Monitor only final task completion rates without examining intermediate decision points, user interaction patterns, or opportunities for beneficial human intervention during agent conversations
- ☒ Implement multi-stage evaluation tracking decision confidence scores, user correction patterns, intervention effectiveness, and explainability-satisfaction correlations ✓
- ☐ Rely on periodic manual reviews of random conversation samples without systematic tracking of intervention effectiveness, decision transparency, or user trust indicators
- ☐ Collect anonymous usage statistics without capturing specific decision rationales, user feedback on agent explanations, or transparency improvement opportunities for trust building

Show Explanation

Question: 54

CertyIQ

A development team is building an AI agent capable of autonomously planning and executing multi-step tasks while retaining context and learning from past interactions.
Which practice is most important to enable the agent to effectively manage long-term memory and complex tasks?

- ☐ Use basic rule-based decision methods that emphasize fast responses over adaptive planning.
- ☐ Apply short-term memory approaches that handle each interaction independently of previous ones.
- ☒ Implement memory mechanisms for context retention and apply chain-of-thought prompts to enhance reasoning. ✓
- ☐ Reduce planning features and memory management to keep the system streamlined.

Show Explanation

Question: 55

CertyIQ

You are implementing a RAG (Retrieval-Augmented Generation) solution.
What is the primary purpose of implementing semantic guardrails within a RAG system?

- ☐ To filter out all queries containing specific keywords that have been flagged as problematic.
- ☐ To automatically translate all LLM responses into multiple languages for improved user comprehension.
- ☐ To eliminate all potential harmful entries from the vector database.
- ☒ To establish rules and constraints based on the meaning of user queries and generated responses. ✓

Show Explanation

Question: 56

CertyIQ

An agent is tasked with solving a series of complex mathematical problems that require external tools to find information. It often struggles to keep track of intermediate steps and reasoning.
Which prompting technique would be MOST effective in improving the agent's clarity and reducing errors in its reasoning?

- ☒ ReAct ✓
- ☐ Symbolic Planning
- ☐ Zero-shot CoT
- ☐ Multi-Plan Generation

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Question: 57

CertyIQ

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Question: 58

CertyIQ

When evaluating a customer service agent's resilience to API failures and network issues, which analysis methods effectively identify weaknesses in error handling and retry mechanisms?
Pick the 2 correct responses below

- ☒ Conduct failure injection testing with varied error types (timeouts, rate limits, malformed responses) while monitoring recovery patterns and fallback behavior. ✓
- ☐ Implement retry mechanisms that standardize recovery attempts across scenarios, emphasizing consistency in handling errors.
- ☒ Analyze retry logic for exponential backoff patterns, retry limits, and circuit breaker integration to prevent cascading failures in distributed systems. ✓
- ☐ Use fixed retry intervals to avoid the pitfalls of dynamic tuning, keeping retry timing consistent across different error conditions.
- ☐ Test under normal network conditions to establish baseline behavior, comparing results against production performance during degraded service scenarios.

Show Explanation

Question: 59

CertyIQ

You're evaluating the performance of a tool-using agent (e.g., one that issues API calls or executes functions). From the list below, what are two important features to evaluate?
Pick the 2 correct responses below

- ☒ Tool use accuracy ✓
- ☐ Tool use rate
- ☒ Task completion rate ✓
- ☐ Tokens per second

Show Explanation

Question: 60

CertyIQ

An AI Engineer at a retail company is developing a customer support AI agent that needs to handle multi-turn conversations while keeping track of customers' previous queries, preferences, and unresolved issues across multiple sessions. Which approach is most effective for managing context retention and enabling the agent to respond coherently in real time?

- ☐ Use a sliding window of recent conversation tokens in memory to track only the last few exchanges.
- ☐ Retrain the model periodically using historical logs to improve long-term contextual understanding.
- ☒ Implement a hybrid memory system with vector-based search and key-value storage to retrieve relevant past interactions. ✓
- ☐ Increase the maximum context window size so the full conversation history is processed each time.

Show Explanation

Question: 61

CertyIQ

When analyzing an agent's failure to complete multi-step financial analysis tasks, which evaluation approach best identifies prompt engineering improvements needed for reliable task decomposition and execution?

- ☐ Focus primarily on response speed optimization as a primary focus over reasoning quality, step completion accuracy, and prompt clarity for complex analytical requirements.
- ☐ Test only final output accuracy as this will automatically include intermediate reasoning steps, decomposition quality, and prompt structure effectiveness for complex workflows.
- ☐ Rely on generic prompt templates which are by default already optimized for general use, instead of tailoring them to financial terminology, calculation needs, or specialized multi-step analysis patterns.
- ☒ Implement systematic prompt testing with chain-of-thought reasoning templates, step-by-step decomposition analysis, and success rate tracking across tasks of varying complexity. ✓

Show Explanation

Question: 62

CertyIQ

Your team has built an agent using LangChain and needs to implement guardrails for deployment in a production environment. Which approach represents the MOST effective integration of NVIDIA NeMo Guardrails?

- ☒ Wrap the LangChain agent with NeMo Guardrails configuration while maintaining the existing workflow architecture and preserving current development investments. ✓
- ☐ Run the LangChain agent in parallel with NeMo Guardrails, allowing comparison of outputs between systems for comprehensive safety validation and performance optimization.
- ☐ Rebuild the agent using only NeMo Guardrails, thereby reconstructing the LangChain implementation with enhanced safety controls and production-ready guardrail integration.
- ☐ Configure input filtering to address safety requirements, integrating guardrail mechanisms focused on data validation and moderation within the current framework.

Show Explanation

Question: 63



When analyzing user feedback patterns to improve a technical documentation agent, which evaluation methods effectively translate feedback into actionable optimization strategies?
Pick the 2 correct responses below

- ☐ Collect broad user feedback as-is, enabling rapid accumulation of suggestions and diverse perspectives for potential future analysis.
- ☒ Implement feedback categorization systems grouping issues by type (accuracy, clarity, completeness) with quantitative impact scoring and improvement prioritization matrices ✓
- ☐ Incorporate user suggestions rapidly to maximize responsiveness and demonstrate continuous adaptation to evolving user needs.
- ☒ Design iterative feedback loops with version tracking, A/B testing of improvements, and regression monitoring to ensure changes enhance rather than degrade performance ✓

Show Explanation

Question: 64



When designing tool integration for an agent that needs to perform mathematical calculations, web searches, and API calls, which architecture pattern provides the most scalable and maintainable approach?

- ☒ Microservice-based tool architecture with standardized interfaces ✓
- ☐ Monolithic tool handler with conditional logic for different tool types
- ☐ External tool services with manual configuration for each agent instance
- ☐ Embedded tool functions within the main agent code

Show Explanation

Question: 65



When designing complex agentic workflows that include both sequential and parallel task execution, which orchestration pattern offers the greatest flexibility?

- ☐ Linear pipeline orchestration with a fixed task sequence
- ☐ Event-driven orchestration that triggers tasks reactively, in series or in parallel
- ☒ Graph-based workflow orchestration incorporating conditional branches ✓

Show Explanation

Question: 66



You are designing a virtual assistant that helps users check weather updates via external APIs. During testing, the agent frequently calls the incorrect tools, often hallucinating endpoints or returning incorrect formats. You suspect the prompt structure might be the root cause of these failures.
Which prompt design best supports consistent tool invocation in this agent?

- ☐ Include tool names in natural language but without parameter examples
- ☐ Rely on the agent's internal knowledge to infer tool usage
- ☐ Provide only a generic system instruction with no examples
- ☒ Use structured prompt templates with few-shot tool usage examples ✓

Show Explanation

Question: 67



Your team has deployed a generative agent for internal HR use, including summarizing candidate resumes and suggesting interview questions. After deployment, you've noticed that the model occasionally associates certain names or genders with particular roles. Which mitigation strategy is the most effective and scalable for reducing this type of bias in agent outputs?

- ☐ Randomly replace names in prompts to reduce identity correlation
- ☒ Implement guardrails to prevent outputs referencing protected attributes ✓
- ☐ Add more training examples to the training dataset and re-train the model
- ☐ Adjust system prompts to explicitly instruct the agent to avoid assumptions based on demographic features

Show Explanation

Question: 68



A financial services agentic AI is being used to automate initial customer onboarding. The agent is completing the process efficiently and accurately, but reviews of its conversations reveal it often uses overly formal and complex language that confuses customers. Which type of evaluation is best suited to address this issue?

- ☒ Controlled user testing sessions to collect user feedback on the clarity and tone of responses ✓
- ☐ Compliance review of the agent's access to regulatory guidelines and policy documentation
- ☐ Continuous user feedback collection, specifically gathering subjective assessments of the agent's communication style
- ☐ Statistical analysis of the agent's decision-making patterns to detect overly formal and complex response choices

Show Explanation

Question: 69



In your RAG deployment, you've identified a performance bottleneck in the retrieval phase – specifically, the time it takes to access the vector database. Which of the following optimization strategies is most aligned with micro-service best practices, considering your RAG architecture?

- ☒ Introduce a dedicated service responsible solely for querying the vector database and returning relevant chunks. ✓
- ☐ Implement a "cache-and-check" mechanism where the retrieval microservice immediately returns the first matching chunk, regardless of relevance.
- ☐ Increase the size of the LLM model itself, because it will automatically accelerate the overall response time.
- ☐ Optimize the LLM prompt to be shorter and more concise, significantly reducing the computational load.

Show Explanation

Question: 70



In a global financial firm, an AI Architect is building a multi-agent compliance assistant using an agentic AI framework. The system must manage short-term memory for multi-turn interactions and long-term memory for persistent user and policy context. It should enable contextual recall and adaptation across sessions using NVIDIA's tool stack. Which architectural approach best supports these requirements?

- ☐ Leverage NVIDIA Triton Inference Server with dynamic batching to cache session-level inputs between inference calls, and use an external Redis store for long-term memory.
- ☒ Leverage NVIDIA NeMo Framework with modular memory management, integrating conversational state tracking, knowledge graphs, and vector store retrieval, while using LoRA-tuned models to adapt responses over time. ✓
- ☐ Leverage RAPIDS cuDF for memory tracking by streaming multi-turn conversation logs as GPU-resident data frames, assuming transactional history can be recalled and reasoned over using dataframe operations.

- ☐
- Rely exclusively on TensorRT to encode all prior knowledge into compiled model weights, allowing inference-only execution with no external memory dependencies across sessions.

Show Explanation

Question: 71

CertyIQ

A technology startup is preparing to launch an AI agent platform to serve clients with unpredictable usage patterns. They face periods of high user activity and low demand, so their deployment approach must minimize wasted resources during slow times and automatically allocate more resources during busy periods — all while keeping operational costs reasonable. Given these requirements, which deployment strategy most effectively ensures both cost-effectiveness and adaptability for scaling agentic AI systems?

- ☒ Implementing autoscaling policies in a container orchestration environment to automatically adjust resources according to workload changes ✓
- ☐ Monitoring system logs for usage patterns and making infrastructure changes after monthly analysis
- ☐ Scheduling periodic manual reviews to increase or decrease infrastructure based on predicted user numbers
- ☐ Using fixed-size virtual machine clusters to guarantee consistent resource allocation at all times

Show Explanation

Question: 72

CertyIQ

When analyzing safety violations in a financial advisory agent that uses NeMo Guardrails, which evaluation approach best identifies gaps in guardrail coverage?

- ☒ Analyze violation patterns, test adversarial prompts, measure guardrail activation, and align policies with observed failures. ✓
- ☐ Monitor overall guardrail activations and system logs to assess operational behavior across different interaction types.
- ☐ Conduct functional testing with representative user inputs to verify policy enforcement in typical usage scenarios.
- ☐ Apply keyword and rule-based validation methods to confirm compliance with policy terms and common risk conditions.

Show Explanation

Question: 73

CertyIQ

Your agent's primary task is to collect data from a device management platform API. This API frequently returns data in JSON format, but the underlying schemas can evolve without detailed documentation. What's the most effective strategy for the agent to maintain consistent access to the data, despite potential schema changes?

- ☐ Building a fixed data extraction method based on the API's historical response patterns.
- ☐ Building a JSON schema validator that constantly monitors and enforces the current API schema.
- ☐ Manually inspecting the API response's structure and updating the agent's code when a change is detected.
- ☒ Building a flexible data mapping layer that can adapt to changes in the API response structure. ✓

Show Explanation

Question: 74

CertyIQ

A Lead AI Architect at a global financial institution is designing a multi-agent fraud detection system using an agentic AI framework. The system must operate in real time, with distinct agents working collaboratively to monitor and analyze transactional patterns across accounts, retain and share contextual information over time, and escalate suspicious behaviors to a human fraud analyst when needed. Which architectural approach enables intelligent specialization, shared memory, and inter-agent coordination in a dynamic and evolving threat environment?

- ☐ Design an agentic workflow where each agent acts independently on isolated data slices with no inter-agent communication to reduce latency and model complexity.
- ☒ Design a modular multi-agent system where individual agents collaborate asynchronously using shared memory and structured messaging. ✓
- ☐ Design a multi-agent system where individual agents collaborate synchronously using shared memory and structured messaging.
- ☐ Design monolithic LLM-based agents that handle all fraud detection tasks within a single loop, without modular roles or multi-agent coordination.
- ☐ Design a centralized rule-based service that checks all transactions against static fraud indicators and sends alerts when thresholds are exceeded.

Show Explanation

Question: 75

CertyIQ

You are evaluating your RAG pipeline. You notice that the LLM-as-a-Judge consistently assigns high similarity scores to responses that contain irrelevant information. What should you investigate as the most likely potential cause with the least development effort?

- ☐ The quality of the synthetic questions used for evaluation.
- ☐ The size of the knowledge base used to power the RAG pipeline.
- ☒ The prompt used to instruct the LLM-as-a-Judge to assess the response. ✓
- ☐ The temperature setting used by the LLM during response generation.

Show Explanation



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